Deakin University
Faculty of Science, Engineering and Built Environment
School of IT

Assignment 1 Report

SIT215 – Computational Intelligence

Trimester 1, 2026

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Tasks Completed	Completed (✓)	Notes (Optional)
Task 1: Environment modelling and problem formulation	☐	
Task 2: Search-based agent implementation and comparison • A^* • Selected uninformed algorithm	☐	
Task 3: Success metrics and evaluation	☐	
Task 4: New heuristic design and performance	☐	

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1. Problem Formulation

1.1 Problem Context

Vingroup JSC [1] is a multi–sector corporation that has thrived as one of the top conglomerates in Vietnam. Since 2000s, it has expanded across different domains, e.g. VinFast - multinational automotive manufacturer, VinAI - leading AI Research company, and many more. It is now aiming to open a new subsidiary, namely VinExpress, focusing on courrier, express, and parcels service market, which is a profitable sector, but has been predominated [2] by ViettelPost, VietnamPost, and J&T Express.

Prior to expanding the market nationally and potentially on a global scale, it first demands a comprehensive efficient route planning in Ho Chi Minh City (HCMC), the largest metropolitan city in Vietnam. Therefore, this report focuses on planning in a small part of the city to evaluate the feasibility and competitiveness when scaling to the whole HCMC. The map is given below.

Figure 1: A snapshot of a selected region of HCMC with 21 landmarks adapted from GoogleMaps [3].

1.2 Problem Constraints and Environment Selection Reasoning

As shown in the map, it is a densely populated residential area, which has imposed pressure on transport infrastructure and led to serious traffic congestion and poorly maintained streets [4]. Express delivery companies must propose appropriate measures to efficiently allocate budget for transportation and management while protecting fragile parcels and on–time delivery. Moreover, planners must be aware of vehicle restrictions in specific roads, traffic flow at different time periods, one–way roads, and limited road widths. Under this context, route feasibility is attributed to two **dynamic constraints**, i.e. vehicle restrictions and traffic flow, and several **static constraints** relating to roads, e.g. directions and road conditions.

As depicted above, the 21 selected landmarks are subject to spread across the map, so that some are near to the avenues, whereas some are located within inner and smaller streets. Besides, distances between landmarks vary considerably. Each path segment is assigned to a weight, which represents projected petrol costs considering traffic flow and distance. This is more reasonable and realistic than using distance, as time taken to travel in some roads may be more or less than expected due to traffic

situations. Fortunately, road fees can be neglected as there are no toll stations in the central of HCMC.

2. Modelling Approach

2.1 PEAS Framework

The problem statement above can be summarised more concisely by the utilisation of PEAS framework [5] to formally define the rationality standards [6] of truck delivery agent.

Environment (E)

- Partially observable (blind spots of truck)
- Multi–agent (other vehicles)
- Nondeterministic (accidents)
- Sequential (braking may cause late delivery)
- Dynamic (traffic flows vary)
- Continuous (time and actions, e.g. speed, rotatation are continuous)
- Known (truck must follow the designated routes)

Performance (P)

- Fuel cost optimisation
- Timely delivery
- No items are broken during transportation
- Right items, right places

Sensors (S)

- Traffic observations
- Clock
- GPS location
- Delivery status

Actuators (A)

- Accelerate
- Brake
- Steering

2.2 Environment Abstraction into State Space: A relaxed version

Having studied the PEAS framework above, it is apparent that the problem context is far from simplicity. Although it reflects the real–world scenarios well, it is arduous to implement an effective and efficient agent in such environment. Therefore, in this work, we would relax the problem by maintaing the static constraints while reducing dynamic constraints into static constraints.

2.3 Goal State and Feasibility

2.4 State Representation

2.5 Successor Function

2.6 Cost Model

Given a graph $G = (V, E)$, for an edge $(u, v) \in E$. The corresponding weight, denoted as $\Omega(u, v)$, given by:

$$\omega(u, v) = \alpha \cdot d(u, v) + \beta \cdot t(u, v) \tag{1}$$

$$\Omega(u, v) = (1 + \tau(u, v) + \psi(u, v)) \times \omega(u, v) \tag{2}$$

where

- $d(u, v)$: estimated distance (in km) adapted from GoogleMaps [3]
- α: fuel comsumption cost
- $t(u, v)$: estimated time (in hour) adapted from GoogleMaps [3]
- $\beta \in (0, 1)$: time contribution coefficient
- $\tau(u, v) \in (0, 1)$: risk factor represents road condition
- $\psi(u, v) \in (0, 1)$: traffic congestion rate

2.7 Relaxed Version of the Problem

2.8 Justification of Modelling Choices

3. Search Strategies and Heuristic Design

3.1 Benchmark Algorithm: A*

3.2 Baseline Heuristic: Manhattan Distance

3.3 Selected Uninformed Search Algorithm

3.4 Comparison Rationale

3.5 Heuristic Properties

3.6 New Heuristic Design

3.7 Expected Impact of the New Heuristic

4. Implementation Overview

4.1 Overall System Structure

4.2 Data Structures

4.3 Node Expansion Logic

4.4 Constraint Validation Logic

4.5 Implementation of A*

4.6 Implementation of the Uninformed Search Algorithm

4.7 Implementation of the New Heuristic

4.8 Code Snippet References

Listing 1: Placeholder for annotated code snippet with line numbers.

```
# Paste relevant code snippet here
# Make sure line numbers are visible in the PDF output
```

Listing 2: A* Search Algorithm

```
def astar(graph, start, goal, heuristic):
    open_set = PriorityQueue()
    open_set.put((0, start))

    came_from = {}
    g_score = {start: 0}

    while not open_set.empty():
        _, current = open_set.get()

        if current == goal:
            return reconstruct_path(came_from, current)

        for neighbor, cost in graph[current]:
            tentative = g_score[current] + cost

            if neighbor not in g_score or tentative < g_score[neighbor]:
                came_from[neighbor] = current
                g_score[neighbor] = tentative
                f_score = tentative + heuristic(neighbor, goal)
                open_set.put((f_score, neighbor))
```

5. Results and Evaluations

5.1 Experimental Setup

5.2 Test Cases

Case	Start Node	Goal Node	Notes
1			
2			
3			

Table 1: Start-goal test cases used for evaluation.

5.3 Performance Metrics

5.4 Comparison of A* and Uninformed Search

Case	Algorithm	Path Found	Path Cost	Nodes Expanded	Runtime	Optimal?
1	A*					
1	Uninformed					
2	A*					
2	Uninformed					
3	A*					
3	Uninformed					

Table 2: Performance comparison of A* and the selected uninformed search algorithm.

5.5 Evaluation of Intelligence Quality

5.6 Comparison of Heuristics

Case	Heuristic	Path Cost	Nodes Expanded	Runtime	Comments
1	Manhattan				
1	New heuristic				
2	Manhattan				
2	New heuristic				
3	Manhattan				
3	New heuristic				

Table 3: Comparison between Manhattan distance and the new heuristic.

5.7 Visualisations

Figure 2: Placeholder for result visualisation (e.g., runtime or node expansion comparison).

5.8 Discussion

6. Conclusion

Acknowledgement

References

[1] Vingroup, "About vingroup," n.d. Accessed: 2026-03-18.

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[3] Google, "Google maps," 2026. Accessed: 2026-03-18.

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[5] Deakin University, "Sit215: Computational intelligence." Unit materials, Deakin University, 2026. Accessed: 2026-03-18.

[6] S. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach*. Pearson Education Limited, 4 ed., 2021. Global Edition, Accessed: 2026-03-18.